

Data for the 9-bus system

1. Topology

The topology of 9-bus system is shown in Fig. 1

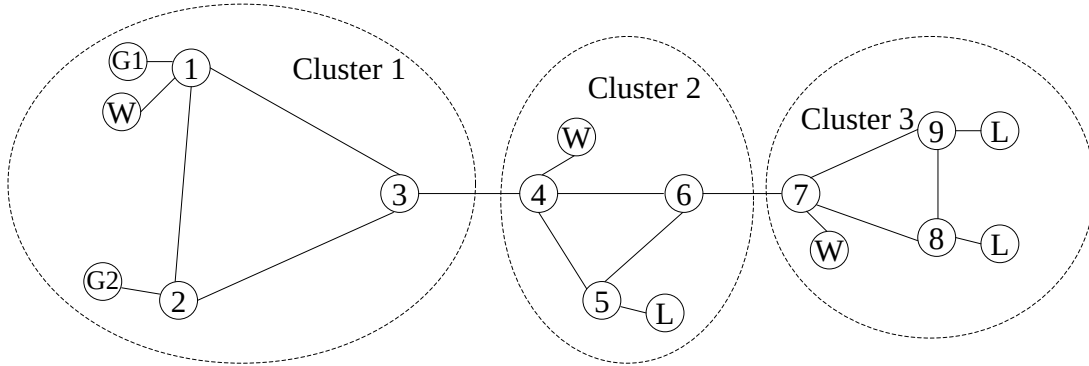


Fig. 1. Topology for the 9-bus system

The traditional units are at bus 1 and 2. The loads are at bus 5, 8 and 9. The wind power is injected at bus 1 4 and 7. Bus 3, 4 and 7 are regarded as the interface of cluster 1, 2 and 3 respectively, so they can be regarded as generators or loads with power buy and sell price depending on specific operation condition.

Specifically, in subcase A and subcase B, the equivalent load at Bus 3 is assumed to be a constant which is 100MW.

2. Data

The power buy and sell price are shown in Table I.

The load shedding and wind spillage cost are shown in Table II.

The network data is shown in Table III.

The generator data are given in Table IV.

The load data are given in Table V.

The wind data available at DS1 are given in Table VI.

The wind data available at DS2 are given in Table VII.

The wind data available at DS3 are given in Table VIII.

In Table VI, VII, and VIII, (N, M) represents the mean and variance of the prediction.

TABLE I
POWER BUY AND SELL PRICE

	DS 1		DS 2		DS 3		Startup cost (\$/MW-hr)
	Buy(\$/MW-hr)	Sell(\$/MW-hr)	Buy(\$/MW-hr)	Sell(\$/MW-hr)	Buy(\$/MW-hr)	Sell(\$/MW-hr)	
Bus 1	5	2	6	1	10	0.8	1.5
Bus 2	3	1.5	20	1	30	0.7	1.5
Bus 3	7	1.5	9	1	20	0.8	N/A
Bus 4	4	1	6	0.8	10	0.5	N/A
Bus 6	6	1.5	9	0.7	13	0.6	N/A
Bus 7	3.5	0.8	5	0.6	9	0.5	N/A

TABLE II
LOAD SHEDDING AND WIND SPILLAGE COST

Load @ bus5	Load @ bus8	Load @ bus9
90 \$/MW-hr	100 \$/MW-hr	100 \$/MW-hr
Wind spillage @ bus1	Wind spillage @ bus4	Wind spillage @ bus7
1 \$/MW-hr	1 \$/MW-hr	1 \$/MW-hr

TABLE III
NETWORK DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	2	0.3	80		5	6	0.12	50
1	3	0.4	80		6	7	0.2	50
2	3	0.3	70		7	8	0.15	30
3	4	0.2	80		7	9	0.15	30
4	5	0.3	50		8	9	0.12	30
4	6	0.2	40					

TABLE IV
GENERATOR DATA

	P _{min}	P _{max}	p ^D	p ^U
Unit 1	15 MW	60 MW	-20 MW/hr	20 MW/hr
Unit 2	10 MW	40 MW	-15 MW/hr	15 MW/hr

TABLE V
LOAD DATA

	DP1	DP2	DP3
Load @ bus5 (MW)	90	87	81
Load @ bus8 (MW)	10	10	7
Load @ bus9 (MW)	5	2	2

TABLE VI
WIND DATA PREDICTED AT DS1

	Prediction at DS1			Conditional prediction at DS2			Conditional prediction at DS3		
	DP1	DP2	DP3	DP1	DP2	DP3	DP1	DP2	DP3
Wind@ bus1 (MW)	(60,3)	(55,3)	(60,3)	(-2,1)	(-2,1)	(-2,1)	(-1,0.5)	(-1,0.5)	(-1,0.5)
Wind@ bus4 (MW)	(10,2)	(9.7,2)	(11.3,2)	(-2.5,1)	(-2.5,1)	(-2.5,1)	(0,1)	(0,1)	(0,1)
Wind@ bus7 (MW)	(2.5,1)	(2.4,1)	(2.4,1)	(2,0.5)	(2,0.5)	(2,0.5)	(1.5,0.5)	(1.5,0.5)	(1.5,0.5)

TABLE VII
WIND DATA PREDICTED AT DS2

	Prediction at DS2			Conditional prediction at DS3		
	DP1	DP2	DP3	DP1	DP2	DP3
Wind@ bus1 (MW)	(55,1)	(53,1)	(53,1)	(-1,0.5)	(-1,0.5)	(-1,0.5)
Wind@ bus4 (MW)	(8.5,1)	(8.3,1)	(8.5,1)	(2,0.5)	(2,0.5)	(2,0.5)
Wind@ bus7 (MW)	(4.5,0.8)	(4,0.8)	(4.4,0.8)	(1.5,0.5)	(1.5,0.5)	(1.5,0.5)

TABLE VIII
WIND DATA PREDICTED AT DS3

	Prediction at DS3		
	DP1	DP2	DP3
Wind@ bus1 (MW)	(52.81,0)	(50.03,0)	(50.67,0)
Wind@ bus4 (MW)	(11.7,0)	(10.4,0)	(9.7,0)
Wind@ bus7 (MW)	(6.8,0)	(7.3,0)	(7.3,0)

Data for the Gansu power grid in China

1. Topology

The topology of the Gansu power grid in China is shown in Fig. 2.

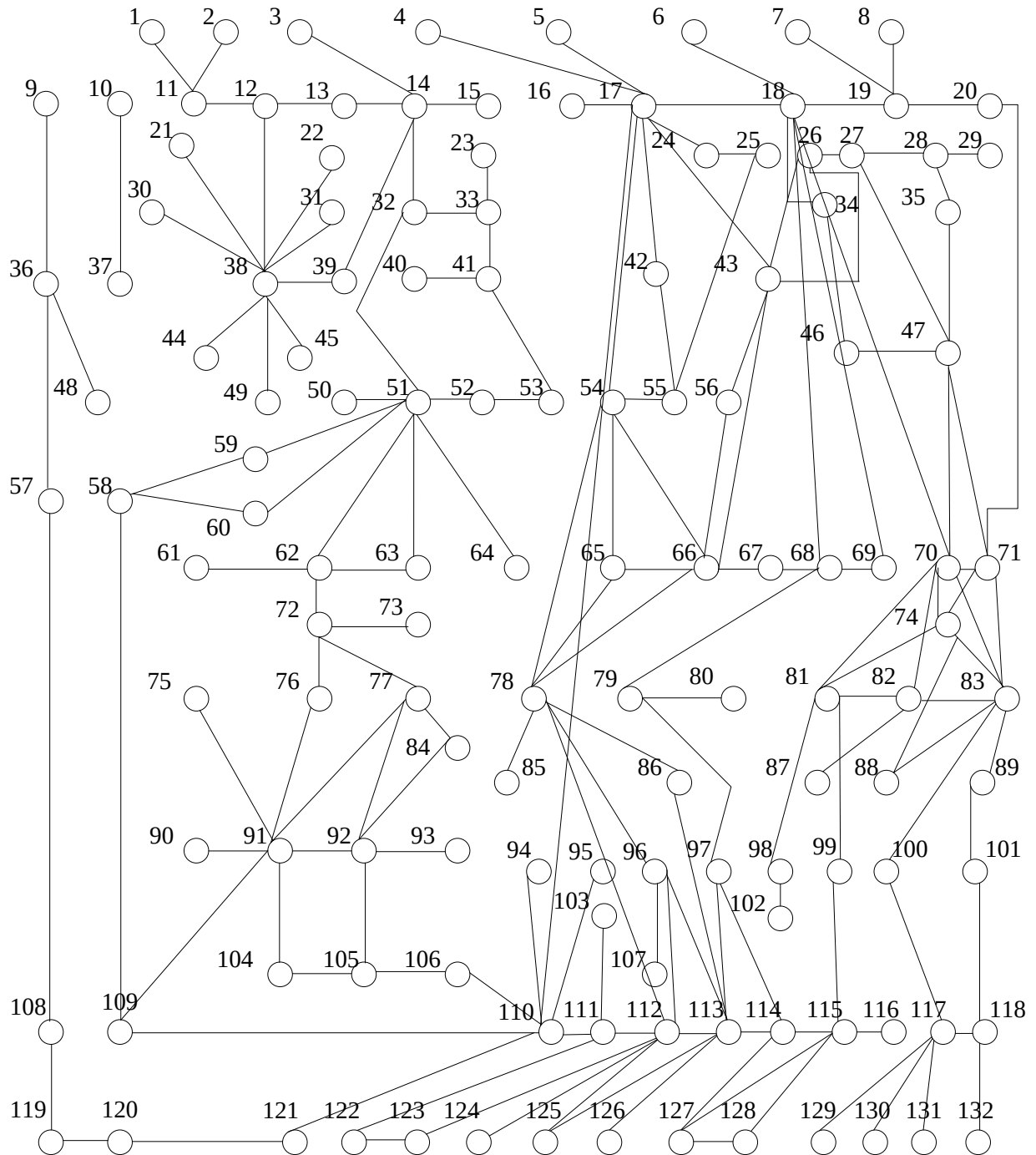


Fig. 2 Topology for the Gansu power grid in China

2. Data

There are 132 buses and 180 transmission lines in the system. There are 25 equivalent traditional generators and 7 wind power generators, injected in bus 14, 32, 38, 39, 51, 62, and 64. The data of generator, generator cost, loads and transmission lines are given in Table IX-Table XII respectively.

TABLE IX
GENERATOR DATA

Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)
4	250	150	±150	43	70	15	±30	87	40	10	±15
6	15	5	±8	45	40	5	±15	90	10	0	±5
7	45	5	±20	50	20	5	±10	91	20	5	±10
8	45	5	±20	46	20	5	±10	114	10	0	±5
10	250	150	±150	61	60	10	±20	121	10	0	±5
16	60	10	±20	69	30	5	±10	131	20	5	±10
20	30	5	±10	73	30	5	±10	132	20	5	±10
23	60	10	±20	80	30	5	±15	N/A	N/A	N/A	N/A
40	45	5	±20	85	30	0	±15	N/A	N/A	N/A	N/A

TABLE X
GENERATOR COST DATA

Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)	Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)
4	10.83	15.46	18.74	2.3	61	10.27	13.42	15.77	2
6	10.32	13.39	15.33	1.2	69	8.74	9.12	12.61	1.8
7	9.78	12.94	15.23	1.5	73	14.49	15.05	18.36	1.8
8	8.47	10.88	12.78	2.1	80	10.05	12.72	13.41	1.4
10	11.03	12.03	13.29	2.5	85	8.35	10.17	11.42	1.2
16	8.97	13.47	14.37	1.7	87	8.82	10.43	11.59	1.9
20	15.43	15.9	16.07	1.9	90	14.94	15.86	16.99	2.1
23	4.33	7.42	10.41	1.7	91	15.43	16.65	17.63	1.1
40	7.69	8.03	10.07	2.2	114	15.67	18.2	20.04	1.5
43	9.88	10.41	12.02	2.4	121	20.68	22.69	23.87	1.9
45	12.39	13.89	18.38	2.1	131	7.83	9.48	10.2	1.9
50	13.42	13.93	15.24	1.7	132	8.47	9.29	10.57	1.6
46	6.94	7.92	9.17	1.8	N/A	N/A	N/A	N/A	N/A

TABLE XI
LOAD DATA

Bus No.	Load(MW)	Bus No.	Load(MW)	Bus No.	Load(MW)
1	7.13	40	0	79	1.9165
2	10.24	41	7.2357	80	0
3	7.43	42	4.29	81	12.6094
4	0	43	0	82	21.3294
5	27.9391	44	3.29	83	5.3237
6	0	45	14.94	84	4.72
7	0	46	0	85	0
8	0	47	14.52	86	7.63
9	22.3476	48	15.33	87	0
10	0	49	12.378	88	1.5052
11	4.58	50	0	89	3.5822
12	5.72	51	0	90	0
13	5.37	52	3.0554	91	0
14	2.29	53	20.422	92	2.3161
15	9.53	54	5.8447	93	0
16	0	55	6.9477	94	22.8798
17	2.34	56	7.8794	95	23.9808
18	18.28	57	40.9793	96	2.6539
19	17.7797	58	36.83	97	2.94
20	0	59	0	98	0.5148
21	5.92	60	0	99	6.7314
22	7.22	61	0	100	4.73
23	0	62	6.0952	101	6.95
24	2.8426	63	8.3619	102	8.2794
25	3.28	64	4.75	103	5.21
26	2.21	65	3.7896	104	3.532
27	9.3985	66	7.3182	105	3.18
28	2.4302	67	18.2274	106	7.8261
29	50.23	68	7.4661	107	18.43
30	7.3	69	0	108	50.82
31	2.489	70	0	109	64.3852
32	2.84	71	6.0774	110	42.29
33	5.69	72	22.9517	111	12.02
34	7.68	73	0	112	4.1494
35	9.87	74	5.1316	113	2.48
36	14.6856	75	2.43	114	0
37	9.6856	76	12.5747	115	5.67
38	5.71	77	3.8423	116	4.57
39	2.56	78	2.4201	117	3.47

TABLE XI
LOAD DATA (CONTINUED)

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
118	3.7815		123	3.49		128	1.4446
119	30.79		124	1.13		129	0
120	32.9078		125	1.59		130	0
121	52.0279		126	0		131	0
122	2.36		127	1.4833		132	0

TABLE XII
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	11	0.47	20		25	55	0.1	20
2	11	0.107	20		26	34	0.1	20
3	14	0.146	20		26	27	0.1	80
4	17	0.3	300		27	28	0.143	80
5	17	0.3	50		27	47	0.143	20
6	18	0.3	80		28	35	0.143	20
7	19	0.3	80		28	29	0.143	80
8	19	0.3	80		30	38	0.95	20
9	36	0.13	80		31	38	0.95	20
10	37	0.5	220		32	33	0.95	50
10	11	0.5	150		32	51	0.95	50
11	12	0.5	80		33	41	0.1	80
12	38	0.2	80		34	46	0.46	20
12	13	0.3	30		17	54	0.1	80
14	15	0.5	20		35	47	0.215	50
14	39	0.5	30		36	57	0.1	120
14	32	0.5	30		36	37	0.1	180
14	16	0.5	50		36	48	0.1	30
16	17	0.1	80		37	58	0.13	50
17	18	0.15	50		37	38	0.13	80
17	54	0.15	50		38	39	0.11	80
17	42	0.15	20		38	44	0.11	20
17	43	0.15	20		38	45	0.11	30
17	24	0.15	20		38	49	0.17	50
17	110	0.15	300		40	41	0.5	80
18	34	0.12	50		41	53	0.12	80
18	68	0.12	20		42	55	0.12	20
18	69	0.1	50		43	46	0.17	30
18	70	0.1	50		43	66	0.17	50
18	19	0.1	80		46	47	0.22	50
19	20	0.14	50		47	70	0.176	50
20	71	0.15	50		47	71	0.52	20
21	38	0.15	20		50	51	0.6	50
22	38	0.15	20		51	52	0.35	50
22	38	0.15	20		51	59	0.35	30
23	33	0.1	100		51	60	0.326	30
24	25	0.1	20		51	62	0.35	20

TABLE XII
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
51	63	0.47	30		77	91	0.25	20
51	64	0.35	50		77	92	0.25	20
52	53	0.4	50		77	84	0.25	20
54	78	0.4	30		78	85	0.58	50
54	65	0.4	20		78	112	0.58	20
54	66	0.4	20		78	96	0.1	50
54	55	0.4	20		78	86	0.1	50
56	66	0.36	20		79	97	0.39	50
57	108	0.19	80		79	80	0.44	50
58	59	0.15	30		81	98	0.25	20
58	60	0.15	30		81	99	0.5	20
58	109	0.15	80		81	82	0.5	20
61	62	0.8	100		82	87	0.5	50
62	63	0.1	20		82	83	0.25	20
62	72	0.1	80		83	88	0.548	20
65	78	0.118	20		83	100	0.548	20
65	66	0.63	20		83	89	0.11	20
66	67	0.63	20		84	82	0.15	20
66	78	0.63	20		86	113	0.13	20
67	68	0.5	20		89	101	0.548	20
68	79	0.13	20		90	91	0.548	50
68	69	0.41	20		91	92	0.25	20
70	71	0.21	20		91	109	0.25	50
70	81	0.1	50		91	104	0.25	20
70	82	0.1	20		92	105	0.25	20
70	74	0.1	50		92	93	0.25	20
70	83	0.1	50		94	110	0.4	50
71	74	0.1	50		95	110	0.4	50
71	83	0.1	50		96	107	0.4	50
72	73	0.24	50		96	112	0.4	20
72	76	0.1	50		96	113	0.4	20
72	77	0.1	50		97	113	0.4	50
74	81	0.112	20		97	114	0.4	20
74	88	0.112	20		98	102	0.126	20
74	83	0.112	20		99	115	0.42	20
75	91	0.1	20		100	117	0.7	50
76	91	0.25	20		101	118	0.4	50

TABLE XII
BRANCH DATA (CONTINUED 2)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
102	98	0.2	20		113	125	0.93	20
103	111	0.2	20		113	126	0.1	20
104	105	0.1	20		114	115	0.1	20
105	106	0.1	20		114	127	0.4	20
106	110	0.1	20		115	127	0.12	20
108	119	0.13	30		115	116	0.154	20
109	110	0.125	50		115	128	0.154	20
110	111	0.6	50		117	118	0.154	20
110	121	0.25	110		117	129	0.154	20
111	112	0.25	50		117	130	0.154	20
111	122	0.25	20		117	131	0.154	50
112	113	0.194	50		118	132	0.76	50
112	123	0.163	20		119	120	0.11	80
112	124	0.25	20		120	121	0.17	80
112	125	0.25	20		122	123	0.17	20
113	114	0.71	20		127	128	0.246	20

Data for the regional power grid

1. Topology

The topology of the regional power grid in China and the power interchange schedule is shown in Fig. 3. There are totally 647 buses, 913 transmission lines, 130 units and 25 wind farms. The detail data for each cluster are given as follows.

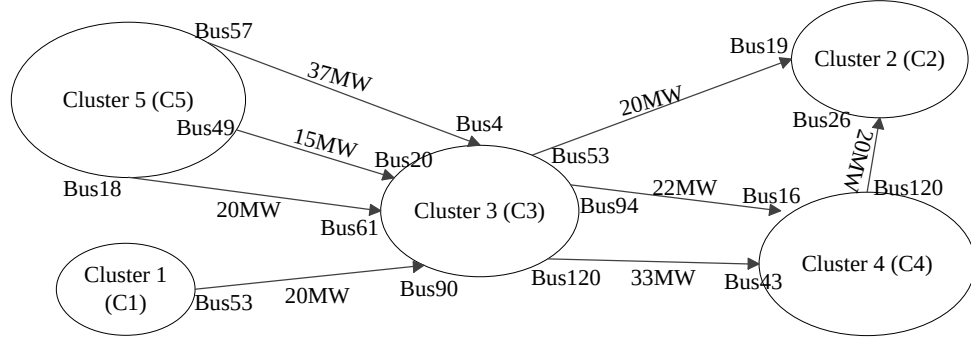


Fig. 3 Topology for the regional power grid in China

2. Data of cluster 1

There are 92 buses and 122 transmission lines in the system. There are 15 equivalent traditional generators and 3 wind power generators, injected in bus 23, 56, and 88. The data of generator, generator cost, loads and transmission lines are given in Table XIII-Table XVI respectively.

TABLE XIII
GENERATOR DATA

Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)	Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)	Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)
3	250	50	± 150	18	50	0	± 30	61	80	0	± 50
7	65	20	± 40	25	80	0	± 50	71	50	10	± 40
8	65	20	± 40	38	65.5	5	± 40	82	40	0	± 30
10	250	50	± 150	43	80	10	± 50	85	40	0	± 30
16	80	20	± 50	48	50	10	± 30	90	40	0	± 30

TABLE XIV
GENERATOR COST DATA

Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)	Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)
3	8.27	9.924	12.4050	2.3	43	5.19	6.228	7.7850	2
7	12.46	14.952	18.6900	1.2	48	7.74	9.288	11.6100	1.8
8	6.98	8.376	10.4700	1.5	61	15.82	18.984	23.7300	1.8
10	5.39	6.468	8.0850	2.1	71	12.66	15.192	18.9900	1.4
16	15.35	18.42	23.0250	2.5	82	8.89	10.668	13.3350	1.2
18	9.03	10.836	13.5450	1.7	85	12.84	15.408	19.2600	1.9
25	13.39	16.068	20.0850	1.9	90	10.74	12.88	16.11	2.1
38	6.52	7.824	9.7800	1.7	N/A	N/A	N/A	N/A	N/A

TABLE XV
LOAD DATA

Bus No.	Load(MW)	Bus No.	Load(MW)	Bus No.	Load(MW)
1	9.43	32	5.84	63	6.32
2	12.63	33	3.69	64	4.75
3	9.88	34	5.68	65	5.78
4	0	35	8.87	66	0
5	30.93	36	16.64	67	20.22
6	2.85	37	5.69	68	15.46
7	10.36	38	7.71	69	0
8	0	39	3.51	70	0
9	20.3400	40	0	71	4.53
10	8.8200	41	9.23	72	25.93
11	0	42	3.29	73	0
12	0	43	0	74	0
13	5.5200	44	0	75	0
14	3.7900	45	14.88	76	15.57
15	10.5300	46	0	77	5.84
16	2.8000	47	13.59	78	8.42
17	0	48	13.32	79	0
18	18.8800	49	10.31	80	0
19	20.7700	50	2.64	81	15.6
20	4.4200	51	0	82	20.83
21	7.9200	52	0	83	4.32
22	0	53	20.42	84	6.72
23	4.4300	54	5.84	85	0
24	2.8400	55	6.94	86	7.63
25	3.2800	56	7.87	87	0
26	0	57	47.97	88	0
27	9.3900	58	30.83	89	4.32
28	0	59	0	90	5.53
29	53.2300	60	0	91	0
30	7.3000	61	0	92	2.75
31	5.3800	62	4.09	N/A	N/A

TABLE XVI
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)	From bus No.	To bus No.	x	Thermal limit(MW)
1	11	0.4	20	26	34	0.12	20
2	11	0.1	20	26	43	0.12	80
3	14	0.1	150	26	27	0.12	80
4	17	0.2	100	27	28	0.14	80
5	17	0.2	50	27	47	0.14	20
6	18	0.23	80	28	35	0.14	20
7	19	0.23	80	28	29	0.14	80
8	19	0.23	80	30	38	0.50	20
9	36	0.1	80	31	38	0.50	20
10	37	0.3	150	32	33	0.50	50
10	11	0.3	150	32	51	0.50	50
11	12	0.3	80	33	41	0.18	80
12	38	0.1	80	34	46	0.30	20
12	13	0.2	30	35	47	0.21	50
13	14	0.2	50	36	57	0.10	120
14	15	0.3	20	36	37	0.10	180
14	39	0.3	30	36	48	0.10	30
14	32	0.3	30	37	58	0.13	50
14	16	0.3	50	37	38	0.13	80
16	17	0.1	80	38	39	0.11	80
17	18	0.15	50	38	44	0.11	20
17	54	0.15	50	38	45	0.11	30
17	42	0.15	20	38	49	0.30	50
17	43	0.15	20	40	41	0.10	80
17	24	0.15	20	41	53	0.20	80
18	34	0.2	50	42	55	0.20	20
18	68	0.2	20	43	46	0.20	30
18	69	0.1	50	43	66	0.20	50
18	70	0.1	50	46	47	0.20	50
18	19	0.1	80	47	70	0.12	50
19	20	0.1	50	47	71	0.37	20
20	71	0.15	50	50	51	0.37	50
21	38	0.15	20	51	52	0.35	50
22	38	0.15	20	51	59	0.35	30
23	33	0.12	100	51	60	0.35	30
24	25	0.12	20	51	62	0.35	50
25	55	0.12	20	51	63	0.41	30

TABLE XVI
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
51	64	0.35	50		71	74	0.12	20
52	53	0.40	50		71	83	0.12	20
54	78	0.40	30		72	73	0.24	50
54	65	0.40	20		72	76	0.10	50
54	66	0.40	20		72	77	0.10	50
54	55	0.40	20		74	81	0.12	20
56	66	0.36	20		74	88	0.12	20
58	59	0.15	30		74	83	0.12	20
58	60	0.15	30		75	91	0.10	20
61	62	0.10	100		76	91	0.17	20
62	63	0.10	20		77	91	0.17	20
62	72	0.10	80		77	92	0.17	20
65	78	0.18	20		77	84	0.27	20
65	66	0.30	20		78	85	0.27	50
66	67	0.30	20		78	86	0.10	50
66	78	0.30	20		79	80	0.35	50
67	68	0.42	20		81	82	0.20	20
68	79	0.13	50		82	87	0.20	50
68	69	0.11	20		82	83	0.15	20
70	71	0.21	20		83	88	0.22	20
70	81	0.12	50		83	89	0.11	20
70	82	0.12	20		84	92	0.15	20
70	74	0.12	20		90	91	0.22	50
70	83	0.12	20		91	92	0.25	20

3. Data of cluster 2

There are 121 buses and 165 transmission lines in the system. There are 22 equivalent traditional generators and 0 wind power generator. The data of generator, generator cost, loads and transmission lines are given in Table XVII-Table XX respectively.

TABLE XVII
GENERATOR DATA

Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)
4	250	100	±125	43	80	10	±40	85	80	15	±40
7	90	15	±45	45	90	10	±45	88	90	15	±45
8	90	15	±45	46	50	5	±25	94	80	0	±40
10	200	100	±100	50	50	5	±25	98	50	0	±25
19	80	10	±40	61	70	5	±35	110	50	0	±25
20	80	0	±40	69	70	10	±35	115	50	0	±25
26	80	0	±40	71	80	20	±40	N/A	N/A	N/A	N/A
42	90	10	±45	83	80	15	±40	N/A	N/A	N/A	N/A

TABLE XVIII
GENERATOR COST DATA

Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)	Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)
4	6.8	8.16	10.2	1.8	50	16.91	20.292	25.365	2.36
7	4.73	5.676	7.095	1.7	61	5.20	6.24	7.8	2.31
8	9.46	11.352	14.19	2.5	69	18.78	22.536	28.17	1.33
10	15.	18.	22.5	2.5	71	20.75	24.9	31.125	1.24
19	6.9	8.28	10.35	1.3	83	16.05	19.26	24.075	2
20	13.48	16.176	20.22	1.9	85	11.31	13.572	16.965	2.92
26	4.3	5.16	6.45	1.8	88	11.38	13.656	17.07	1.68
42	7.6	9.12	11.4	2.2	94	6.94	8.328	10.41	2.17
43	9.85	11.82	14.775	2.4	98	9.4	11.28	14.1	1.45
45	7.34	8.808	11.01	2.5	110	8.66	10.392	12.99	2.5
46	9.46	11.352	14.19	1.5	115	8.66	10.392	12.99	1.51

TABLE XIX
LOAD DATA

Bus No.	Load(MW)	Bus No.	Load(MW)	Bus No.	Load(MW)
1	4.20	40	2.91	79	7.95
2	9.20	41	7.27	80	0
3	5.43	42	4.29	81	10.64
4	2.29	43	0	82	25.34
5	8.73	44	3.90	83	2.37
6	6.22	45	14.40	84	0
7	9.28	46	0	85	1.89
8	2.61	47	14.20	86	5.91
9	2.34	48	15.30	87	0
10	3.82	49	12.38	88	0
11	0	50	2.82	89	0
12	0	51	5.91	90	0
13	0	52	3.04	91	0
14	0	53	20.40	92	10.82
15	10.53	54	5.87	93	0
16	10.53	55	6.97	94	20.88
17	20.04	56	7.84	95	18.98
18	0	57	40.93	96	5.69
19	0	58	0	97	5.90
20	0	59	5.44	98	8.58
21	0	60	2.84	99	3.74
22	9.79	61	2.84	100	9.70
23	1.37	62	6.02	101	7.50
24	2.88	63	8.39	102	3.24
25	3.28	64	4.50	103	5.21
26	2.21	65	7.76	104	1.50
27	5.82	66	2.3820	105	3.10
28	2.43	67	10.24	106	3.81
29	53.81	68	2.41	107	0
30	17.94	69	2.99	108	53.82
31	0	70	0	109	55.32
32	0	71	0	110	30.20
33	0	72	0	111	12.08
34	0	73	0	112	4.14
35	29.87	74	0	113	0
36	14.80	75	2.43	114	2.48
37	9.68	76	2.57	115	5.95
38	5.72	77	9.83	116	10.70
39	2.64	78	7.41	117	2.40

TABLE XIX
LOAD DATA (CONTINUED)

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
118	3.75		120	12.98		N/A	N/A
119	20.79		121	15.09		N/A	N/A

TABLE XX
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	11	0.17	50		25	55	0.25	50
2	11	0.17	50		26	34	0.25	50
3	14	0.17	50		26	43	0.43	80
4	17	0.24	150		26	27	0.13	100
5	17	0.51	80		27	28	0.13	100
6	18	0.41	100		27	47	0.19	80
7	19	0.32	100		28	35	0.27	80
8	19	0.23	100		28	29	0.27	100
9	36	0.16	100		30	38	0.27	70
10	37	0.26	150		31	38	0.27	70
10	11	0.26	150		32	33	0.34	90
11	12	0.26	100		32	51	0.38	90
12	38	0.31	100		33	41	0.38	100
12	13	0.31	80		34	46	0.38	70
13	14	0.49	80		35	47	0.38	90
14	15	0.49	80		36	57	0.42	120
14	39	0.41	30		36	37	0.42	180
14	32	0.42	80		36	48	0.16	80
14	16	0.50	70		37	58	0.16	80
16	17	0.13	100		37	38	0.46	80
17	18	0.13	90		38	39	0.43	80
17	54	0.24	90		38	44	0.33	70
17	42	0.41	80		38	45	0.33	80
17	43	0.25	80		38	49	0.33	80
17	24	0.25	80		40	41	0.33	80
17	110	0.47	150		41	53	0.20	90
18	34	0.11	90		42	55	0.20	90
18	68	0.11	60		43	46	0.20	70
18	69	0.31	90		43	66	0.20	70
18	70	0.24	90		46	47	0.10	80
18	19	0.27	50		47	70	0.10	70
19	20	0.14	50		47	71	0.10	70
20	71	0.15	20		50	51	0.24	70
21	38	0.10	20		51	52	0.24	70
22	38	0.15	20		51	59	0.34	90
23	33	0.24	100		51	60	0.34	90
24	25	0.21	20		51	62	0.50	60

TABLE XX
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
51	63	0.35	60		77	91	0.22	60
51	64	0.35	60		77	92	0.22	60
52	53	0.35	60		77	84	0.22	60
54	78	0.29	60		78	85	0.22	50
54	65	0.29	80		78	112	0.42	60
54	66	0.26	80		78	96	0.42	60
54	55	0.26	80		78	86	0.42	60
56	66	0.29	80		79	97	0.35	60
57	108	0.29	80		79	80	0.35	60
58	59	0.29	60		81	98	0.11	80
58	60	0.14	60		81	99	0.37	80
58	109	0.14	100		81	82	0.37	80
61	62	0.27	100		82	87	0.37	70
62	63	0.27	70		82	83	0.37	70
62	72	0.34	80		83	88	0.20	80
65	78	0.39	70		83	100	0.20	80
65	66	0.39	90		83	89	0.20	70
66	67	0.45	60		84	92	0.40	70
66	78	0.50	60		86	113	0.30	50
67	68	0.45	80		89	101	0.30	50
68	79	0.13	80		90	91	0.40	50
68	69	0.19	60		91	92	0.40	60
70	71	0.19	50		91	109	0.40	80
70	81	0.13	50		91	104	0.30	80
70	82	0.25	70		92	105	0.44	80
70	74	0.25	70		92	93	0.43	80
70	83	0.33	50		94	110	0.27	50
71	74	0.33	40		95	110	0.24	50
71	83	0.38	40		96	107	0.25	70
72	73	0.28	80		96	112	0.27	70
72	76	0.28	80		96	113	0.27	70
72	77	0.28	80		97	113	0.14	50
74	81	0.40	80		97	114	0.14	80
74	88	0.40	80		98	102	0.16	90
74	83	0.40	80		99	115	0.20	90
75	91	0.40	60		100	117	0.22	80
76	91	0.22	60		101	118	0.22	70

TABLE XX
BRANCH DATA (CONTINUED 2)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
102	98	0.31	70		111	112	0.39	80
103	111	0.31	80		112	113	0.39	50
104	105	0.10	80		113	114	0.26	90
105	106	0.10	60		114	115	0.26	70
106	110	0.10	60		115	116	0.27	80
108	119	0.13	60		117	118	0.27	80
109	110	0.12	60		119	120	0.27	60
110	111	0.60	80		120	121	0.27	60
110	121	0.43	110		N/A	N/A	N/A	N/A

4. Data of cluster 3

Cluster 3 is the Gansu power system in China, so the data is the same with the Gansu power system, which have been already given in the above.

5. Data of cluster 4

There are 142 buses and 208 transmission lines in the system. There are 28 equivalent traditional generators and 8 wind power generators, injected in bus 17, 39, 60, 65, 68, 72, 74 and 80. The data of generator, generator cost, loads and transmission lines are given in Table XXI -Table XXIV respectively.

TABLE XXI
GENERATOR DATA

Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)	Bus No.	P _{max} (MW)	P _{min} (MW)	P ^D /P ^U (MW/hr)
5	200	100	±133	48	60	5	±40	126	70	0	±47
6	80	0	±53	65	80	0	±53	128	70	0	±47
7	80	5	±53	70	60	5	±40	134	50	0	±33
8	80	5	±53	75	90	0	±60	135	70	0	±47
15	250	100	±167	79	80	0	±53	136	70	0	±47
16	80	5	±53	85	80	0	±53	137	75	0	±50
18	50	0	±33	87	80	0	±53	139	55	0	±37
25	80	0	±53	95	80	0	±53	140	55	0	±37
37	105.50	5	±70	110	90	0	±60	N/A	N/A	N/A	N/A
43	90	5	±60	115	50	0	±33	N/A	N/A	N/A	N/A

TABLE XXII
GENERATOR COST DATA

Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)	Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)
5	12.31	14.77	18.47	2.01	79	4.47	5.36	6.71	1.1500
6	5.22	6.26	7.83	2.40	85	5.93	7.12	8.90	1.11
7	3.84	4.61	5.76	2.78	87	11.53	13.84	17.29	2.06
8	8.71	10.45	13.07	2.92	95	5.58	6.70	8.37	2.56
15	13.39	16.07	20.09	2.09	110	5.21	6.25	7.82	2.87
16	3.70	4.44	5.55	1.28	115	9.43	11.32	14.15	1.26
18	18.31	21.97	27.46	1.30	126	20.34	24.41	30.51	2.14
25	7.30	8.76	10.95	1.52	128	20.74	24.89	31.11	1.94
37	10.95	13.14	16.42	2.68	134	15.88	19.06	23.82	1.02
43	16.81	20.17	25.21	1.51	135	9.30	11.16	13.95	1.67
48	11.95	14.34	17.92	2.63	136	4.69	5.63	7.04	1.32
65	13.22	15.86	19.83	1.49	137	9.61	11.53	14.42	2.59
70	13.48	16.18	20.22	2.86	139	15.20	18.24	22.80	1.62
75	12.73	15.28	19.10	1.70	140	11.10	13.32	16.65	2.06

TABLE XXIII
LOAD DATA

Bus No.	Load(MW)	Bus No.	Load(MW)	Bus No.	Load(MW)
1	8.13	40	0	79	0
2	12.24	41	10.37	80	0
3	7.42	42	6.69	81	10.60
4	2.18	43	9.72	82	18.32
5	30.93	44	0	83	5.32
6	5.72	45	0	84	0
7	2.64	46	0	85	0
8	9.36	47	10.25	86	9.61
9	10.34	48	10.03	87	0
10	5.24	49	11.06	88	0
11	3.38	50	0	89	5.82
12	6.83	51	0	90	2.94
13	2.38	52	0	91	7.27
14	40.	53	22.51	92	0
15	0	54	0	93	0
16	0	55	12.94	94	20.87
17	0	56	3.87	95	20.04
18	20.82	57	40.03	96	0
19	19.77	58	32.78	97	0
20	4.37	59	0	98	0
21	0	60	0	99	10.73
22	0	61	0	100	0
23	0	62	0	101	15.95
24	12.81	63	5.36	102	9.27
25	5.35	64	0	103	6.35
26	5.74	65	5.78	104	0
27	0	66	4.31	105	0
28	4.29	67	20.22	106	6.38
29	50.23	68	9.46	107	20.33
30	5.93	69	4.92	108	40.97
31	5.42	70	2.46	109	68.38
32	3.47	71	0	110	40.29
33	0	72	18.95	111	0
34	4.92	73	0	112	4.1494
35	8.84	74	0	113	0
36	18.68	75	5.43	114	0
37	2.68	76	15.57	115	10.84
38	7.71	77	10.84	116	6.46
39	0	78	2.42	117	2.44

TABLE XXIII
LOAD DATA (CONTINUED)

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
118	0		127	5.48		136	5.38
119	28.93		128	7.44		137	0
120	30.90		129	0		138	7.90
121	49.02		130	0		139	11.20
122	0		131	2.95		140	1.35
123	0		132	8.83		141	7.20
124	0		133	10.94		142	0
125	0		134	0		N/A	5.38
126	0		135	8.24		N/A	0

TABLE XXIV
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	11	0.18	50		25	55	0.43	50
2	11	0.28	50		26	34	0.43	60
3	14	0.30	50		26	43	0.43	60
4	17	0.30	260		26	27	0.43	80
5	17	0.30	150		27	28	0.13	80
6	18	0.30	80		27	47	0.13	60
7	19	0.30	80		28	35	0.13	70
8	19	0.30	80		28	29	0.13	80
9	36	0.10	80		30	38	0.25	50
10	37	0.10	220		31	38	0.25	50
10	11	0.21	150		32	33	0.15	50
11	12	0.21	80		32	51	0.15	50
12	38	0.21	80		33	41	0.11	80
12	13	0.21	30		34	46	0.14	20
13	14	0.21	50		35	47	0.28	50
14	15	0.35	180		36	57	0.28	120
14	39	0.35	60		36	37	0.28	150
14	32	0.35	60		36	48	0.31	60
14	16	0.35	70		37	58	0.11	60
16	17	0.10	80		37	38	0.11	80
17	18	0.17	50		38	39	0.37	100
17	54	0.17	50		38	44	0.37	70
17	42	0.24	60		38	45	0.37	80
17	43	0.24	60		38	49	0.27	50
17	24	0.17	50		40	41	0.25	80
17	110	0.17	150		41	53	0.12	80
18	34	0.21	50		42	55	0.21	70
18	68	0.21	40		43	46	0.21	70
18	69	0.10	50		43	66	0.21	50
18	70	0.10	50		46	47	0.39	50
18	19	0.10	80		47	70	0.39	50
19	20	0.11	50		47	71	0.44	60
20	71	0.12	50		50	51	0.44	50
21	38	0.12	40		51	52	0.24	50
22	38	0.12	40		51	59	0.24	50
23	33	0.43	150		51	60	0.34	50
24	25	0.43	50		51	62	0.35	40

TABLE XXIV
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
51	63	0.10	50		77	91	0.20	50
51	64	0.10	50		77	92	0.33	80
52	53	0.20	50		77	84	0.33	80
54	78	0.20	70		78	85	0.33	50
54	65	0.20	70		78	112	0.28	50
54	66	0.20	70		78	96	0.28	50
54	55	0.31	70		78	86	0.28	50
56	66	0.31	60		79	97	0.28	50
57	108	0.31	80		79	80	0.31	50
58	59	0.43	60		81	98	0.26	90
58	60	0.27	60		81	99	0.26	70
58	109	0.24	80		81	82	0.29	70
61	62	0.10	100		82	87	0.32	50
62	63	0.10	50		82	83	0.41	40
62	72	0.10	80		83	88	0.45	40
65	78	0.10	80		83	100	0.45	40
65	66	0.10	60		83	89	0.10	80
66	67	0.10	60		84	92	0.10	80
66	78	0.10	60		86	113	0.10	80
67	68	0.10	40		89	101	0.10	80
68	79	0.10	40		90	91	0.10	50
68	69	0.15	40		91	92	0.40	70
70	71	0.15	20		91	109	0.40	50
70	81	0.15	50		91	104	0.40	50
70	82	0.15	40		92	105	0.40	50
70	74	0.15	50		92	93	0.40	50
70	83	0.15	50		94	110	0.33	50
71	74	0.28	50		95	110	0.20	50
71	83	0.28	50		96	107	0.32	50
72	73	0.29	50		96	112	0.32	60
72	76	0.29	50		96	113	0.32	60
72	77	0.29	50		97	113	0.32	50
74	81	0.32	70		97	114	0.39	60
74	88	0.32	70		98	102	0.39	80
74	83	0.32	70		99	115	0.43	80
75	91	0.19	60		100	117	0.43	50
76	91	0.20	60		101	118	0.20	80

TABLE XXIV
BRANCH DATA (CONTINUED 2)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
101	118	0.20	80		120	121	0.30	80
102	98	0.20	60		122	123	0.30	60
103	111	0.20	60		127	128	0.30	60
104	105	0.10	60		133	120	0.2460	60
105	106	0.10	40		133	134	0.16	80
106	110	0.10	40		134	120	0.14	80
108	119	0.10	80		134	135	0.14	40
109	110	0.10	80		134	141	0.14	30
110	111	0.16	80		135	121	0.14	100
110	121	0.26	110		135	142	0.20	60
111	112	0.16	100		136	122	0.20	60
111	122	0.16	80		136	137	0.38	80
112	113	0.39	50		136	142	0.38	80
112	123	0.16	80		137	123	0.38	120
112	124	0.25	50		137	125	0.38	120
112	125	0.25	50		138	126	0.41	50
113	114	0.25	50		138	139	0.10	50
113	125	0.25	70		139	127	0.10	70
113	126	0.31	70		139	128	0.10	40
114	115	0.31	80		139	140	0.18	140
114	127	0.14	60		140	129	0.18	120
115	127	0.17	120		124	142	0.26	120
115	116	0.17	120		122	141	0.26	120
115	128	0.35	120		126	137	0.26	120
117	118	0.35	120		119	141	0.18	120
117	129	0.35	120		117	136	0.18	120
117	130	0.35	120		115	138	0.30	120
117	131	0.35	50		110	141	0.30	120
118	132	0.28	50		107	139	0.30	120
119	120	0.28	50		92	140	0.30	120
77	133	0.3	120		N/A	N/A	N/A	N/A

5. Data of cluster 5

There are 160 buses and 238 transmission lines in the system. There are 40 equivalent traditional generators and 10 wind power generators, injected in bus 21, 25, 42, 46, 60, 61, 68, 75, 78 and 89. The data of generator, generator cost, loads and transmission lines are given in Table XXV -Table XXVIII respectively.

TABLE XXV
GENERATOR DATA

Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)	Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)	Bus No.	$P_{\max}$ (MW)	$P_{\min}$ (MW)	P^D/P^U (MW/hr)
4	250	100	± 167	69	50	5	± 33	137	45	0	± 30
5	25	0	± 17	73	50	0	± 33	139	45	0	± 30
7	55	5	± 37	80	50	0	± 33	140	45	0	± 30
9	55	5	± 37	85	60	0	± 40	144	70	0	± 47
10	250	150	± 167	87	60	0	± 40	145	70	0	± 47
16	80	10	± 53	90	50	0	± 33	146	70	0	± 47
22	70	0	± 47	95	80	0	± 53	148	70	0	± 47
23	80	0	± 53	114	80	0	± 53	149	95	0	± 63
40	85.50	5	± 57	120	80	0	± 53	150	75	0	± 50
43	90	5	± 60	127	80	0	± 53	151	70	0	± 47
45	90	5	± 60	132	50	0	± 33	155	50	0	± 33
50	60	0	± 40	134	50	0	± 33	156	50	0	± 33
46	40	0	± 27	135	50	0	± 33	N/A	N/A	N/A	N/A
65	80	0	± 53	136	40	0	± 27	N/A	N/A	N/A	N/A

TABLE XXVI
GENERATOR COST DATA

Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)		Bus No.	DS 1 (\$/MW-hr)	DS 2 (\$/MW-hr)	DS 3 (\$/MW-hr)	Startup (\$/MW-hr)
4	7.78	9.34	11.67	1.33		95	10.33	12.40	15.50	2.63
5	4.02	4.82	6.03	2.20		114	19.46	23.35	29.19	2.74
7	10.43	12.52	15.65	1.53		120	10.78	12.94	16.17	1.17
9	7.78	9.34	11.67	2.31		127	7.24	8.69	10.86	1.80
10	15.94	19.13	23.91	2.38		132	5.47	6.56	8.21	1.52
16	6.47	7.76	9.71	2.50		134	9.47	11.36	14.21	2.60
22	10.59	12.71	15.89	1.90		135	4.96	5.95	7.44	1.86
23	15.04	18.05	22.56	1.17		136	10.67	12.80	16.01	2.82
40	9.95	11.94	14.93	1.46		137	5.46	6.55	8.19	1.36
43	5.85	7.02	8.77	2.83		139	6.62	7.94	9.93	1.53
45	12.39	14.87	18.59	1.30		140	4.85	5.82	7.28	1.29
50	16.93	20.32	25.40	2.65		144	8.51	10.21	12.77	1.27
46	8.39	10.07	12.59	2.08		145	9.93	11.92	14.90	2.74
65	12.36	14.83	18.54	2.99		146	13.57	16.28	20.36	2.16
69	13.68	16.42	20.52	1.16		148	14.29	17.15	21.44	2.10
73	4.64	5.57	6.96	1.89		149	10.93	13.12	16.40	1.29
80	6.93	8.32	10.40	1.21		150	7.34	8.81	11.01	2.71
85	8.75	10.50	13.13	2.92		151	9.83	11.80	14.75	2.24
87	8.29	9.95	12.43	1.01		155	5.83	7.00	8.75	1.70
90	17.07	20.48	25.61	2.55		156	9.07	10.88	13.61	2.03

TABLE XXVII
LOAD DATA

Bus No.	Load(MW)	Bus No.	Load(MW)	Bus No.	Load(MW)
1	8.23	40	8.82	79	1.00
2	11.25	41	9.2357	80	0
3	8.46	42	0	81	10.60
4	2.79	43	4.29	82	20.32
5	29.93	44	3.29	83	5.32
6	3.48	45	16.94	84	0
7	4.46	46	0	85	0
8	0	47	13.52	86	7.63
9	21.35	48	13.33	87	0
10	0	49	15.37	88	2.50
11	5.59	50	0	89	5.58
12	0	51	2.56	90	0
13	4.32	52	3.05	91	5.25
14	0	53	20.42	92	3.48
15	10.54	54	8.84	93	0
16	0	55	0	94	20.87
17	0	56	0	95	24.98
18	20.26	57	36.97	96	0
19	15.77	58	36.83	97	2.94
20	0	59	3.88	98	2.51
21	6.93	60	0	99	8.73
22	5.22	61	0	100	3.73
23	0	62	7.09	101	5.95
24	5.84	63	7.36	102	8.27
25	1.44	64	3.75	103	5.21
26	3.42	65	0	104	4.53
27	10.39	66	8.31	105	4.18
28	3.43	67	20.22	106	9.82
29	45.23	68	0	107	16.43
30	6.30	69	10.54	108	50.82
31	3.48	70	0	109	65.38
32	3.24	71	6.07	110	45.29
33	0	72	20.95	111	10.02
34	7.35	73	1.46	112	4.14
35	10.46	74	0	113	2.48
36	12.68	75	3.43	114	0
37	11.68	76	10.57	115	6.67
38	6.71	77	4.84	116	3.57
39	0	78	3.42	117	6.47

TABLE XXVII
LOAD DATA (CONTINUED)

Bus No.	Load(MW)		Bus No.	Load(MW)		Bus No.	Load(MW)
118	5.78		132	0		146	22.13
119	25.79		133	8.25		147	21.14
120	30.90		134	5.36		148	0
121	50.02		135	10.24		149	0
122	2.47		136	8.30		150	1.60
123	3.49		137	0		151	5.93
124	1.13		138	7.90		152	4.27
125	1.59		139	11.20		153	0
126	0		140	1.35		154	0
127	1.48		141	7.20		155	6.80
128	1.44		142	0		156	12.80
129	5.82		143	0		157	6.53
130	4.89		144	15.96		158	0
131	0		145	20.38		159	6.90
160	2.84		N/A	N/A		N/A	N/A

TABLE XXVIII
BRANCH DATA

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
1	11	0.37	50		25	55	0.30	30
2	11	0.10	50		26	34	0.30	30
3	14	0.10	50		26	43	0.25	60
4	17	0.10	150		26	27	0.25	100
5	17	0.10	60		27	28	0.14	80
6	18	0.10	70		27	47	0.14	40
7	19	0.10	80		28	35	0.14	40
8	19	0.10	80		28	29	0.13	80
9	36	0.13	80		30	38	0.15	40
10	37	0.30	210		31	38	0.15	40
10	11	0.30	160		32	33	0.15	40
11	12	0.30	80		32	51	0.15	80
12	38	0.30	80		33	41	0.20	80
12	13	0.20	40		34	46	0.16	50
13	14	0.20	50		35	47	0.29	30
14	15	0.10	50		36	57	0.10	120
14	39	0.10	40		36	37	0.10	180
14	32	0.10	40		36	48	0.10	30
14	16	0.10	80		37	58	0.30	60
16	17	0.20	80		37	38	0.30	90
17	18	0.20	60		38	39	0.10	90
17	54	0.10	60		38	44	0.10	50
17	42	0.10	30		38	45	0.10	50
17	43	0.10	30		38	49	0.29	50
17	24	0.10	30		40	41	0.41	80
17	110	0.10	150		41	53	0.20	80
18	34	0.14	50		42	55	0.12	40
18	68	0.14	50		43	46	0.17	40
18	69	0.14	50		43	66	0.17	50
18	70	0.14	50		46	47	0.22	50
18	19	0.15	80		47	70	0.16	50
19	20	0.15	50		47	71	0.51	60
20	71	0.15	50		50	51	0.26	50
21	38	0.15	20		51	52	0.35	50
22	38	0.15	120		51	59	0.35	50
23	33	0.20	100		51	60	0.26	50
24	25	0.20	30		51	62	0.11	40

TABLE XXVIII
BRANCH DATA (CONTINUED 1)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
51	63	0.21	30		77	91	0.25	20
51	64	0.21	50		77	92	0.25	20
52	53	0.21	50		77	84	0.25	20
54	78	0.11	30		78	85	0.35	50
54	65	0.11	40		78	112	0.35	20
54	66	0.14	40		78	96	0.12	40
54	55	0.14	20		78	86	0.12	40
56	66	0.13	20		79	97	0.30	40
57	108	0.19	100		79	80	0.40	40
58	59	0.25	30		81	98	0.25	30
58	60	0.25	30		81	99	0.50	30
58	109	0.25	50		81	82	0.50	30
61	62	0.30	80		82	87	0.50	50
62	63	0.10	30		82	83	0.20	20
62	72	0.10	80		83	88	0.24	20
65	78	0.10	30		83	100	0.24	20
65	66	0.30	30		83	89	0.11	20
66	67	0.30	60		84	92	0.10	20
66	78	0.30	60		86	113	0.13	20
67	68	0.25	60		89	101	0.24	20
68	79	0.13	60		90	91	0.24	50
68	69	0.11	40		91	92	0.24	20
70	71	0.12	40		91	109	0.24	80
70	81	0.10	50		91	104	0.24	60
70	82	0.10	30		92	105	0.24	60
70	74	0.10	30		92	93	0.25	60
70	83	0.10	20		94	110	0.25	30
71	74	0.10	20		95	110	0.25	30
71	83	0.10	30		96	107	0.25	30
72	73	0.24	50		96	112	0.25	50
72	76	0.10	50		96	113	0.25	20
72	77	0.10	50		97	113	0.25	50
74	81	0.12	50		97	114	0.25	20
74	88	0.12	50		98	102	0.12	20
74	83	0.12	30		99	115	0.20	20
75	91	0.10	30		100	117	0.17	50
76	91	0.25	20		101	118	0.14	50

TABLE XXVIII
BRANCH DATA (CONTINUED 2)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
102	98	0.12	20		134	146	0.29	30
103	111	0.12	20		135	121	0.27	100
104	105	0.21	20		135	142	0.21	60
105	106	0.21	20		136	122	0.46	60
106	110	0.21	50		136	137	0.32	80
108	119	0.21	50		136	142	0.38	80
109	110	0.21	40		136	147	0.19	120
110	111	0.16	40		137	123	0.20	120
110	121	0.20	110		137	125	0.20	120
111	112	0.20	60		137	143	0.20	100
111	122	0.20	60		137	158	0.20	100
112	113	0.14	40		138	126	0.14	50
112	123	0.13	40		138	139	0.14	50
112	124	0.35	40		138	143	0.14	70
112	125	0.35	40		139	127	0.14	70
113	114	0.17	20		139	128	0.14	40
113	125	0.30	20		139	140	0.14	40
113	126	0.10	20		139	151	0.17	40
114	115	0.10	20		139	159	0.17	40
114	127	0.24	20		140	129	0.29	120
115	127	0.12	60		140	144	0.21	100
115	116	0.14	20		141	145	0.41	100
115	128	0.14	20		141	146	0.41	100
117	118	0.14	60		142	143	0.41	100
117	129	0.14	20		142	146	0.41	70
117	130	0.14	20		142	147	0.32	70
117	131	0.14	50		142	148	0.32	90
118	132	0.36	50		143	144	0.27	90
119	120	0.11	50		143	149	0.24	90
120	121	0.17	80		143	150	0.31	90
122	123	0.17	20		144	151	0.33	70
127	128	0.26	20		144	152	0.39	70
133	120	0.26	20		144	153	0.39	70
133	134	0.26	80		145	146	0.22	30
134	120	0.26	80		145	154	0.22	30
134	135	0.26	40		145	155	0.22	50
134	141	0.13	30		146	147	0.11	50

TABLE XXVIII
BRANCH DATA (CONTINUED 3)

From bus No.	To bus No.	x	Thermal limit (MW)		From bus No.	To bus No.	x	Thermal limit (MW)
146	155	0.11	50		150	160	0.25	90
147	148	0.31	50		151	152	0.25	90
147	156	0.32	90		152	153	0.43	100
147	157	0.21	90		152	160	0.45	100
148	158	0.21	60		154	155	0.45	90
149	150	0.22	60		155	156	0.33	90
149	159	0.19	50		156	157	0.44	50
149	160	0.19	50		159	160	0.18	50